

Abstract

This paper examines the critical role of digital product designers in creating accessible web applications, sites, and services for all users by delving into concepts and systems covered in these four themes: (1) concepts related to accessibility; (2) the importance of accessibility as a human factors concern and a core responsibility of designers; (3) accessibility gaps and challenges in product design, particularly in startups; and (4) future trends and shifts in the industry regarding accessibility, especially with the growing prevalence of artificial intelligence. Understanding concepts and systems related to accessibility, such as universal design, inclusive design, Web Accessibility Initiative (WAI), and the different disabilities, are crucial for product designers to create accessible products effectively. Learning about design challenges like limitations in executive support and prioritization of initiatives, training, and education support concerning accessibility could help drive more support and an inclusive environment that stresses accessibility as an integral part of the design process, ultimately leading to the creation of inclusive web products and services. Finally, the paper discusses forthcoming innovations in web accessibility through artificial intelligence.

Keywords: Accessibility, disabilities, product design, startups, artificial intelligence

A Focus on Accessibility in Product Design

This paper explores the pivotal role of digital product designers in ensuring that web applications, sites, and services can be experienced to all users. Designing inclusive and accessible products is a goal closely aligned with the principles of human factors, which focus on understanding the interactions between humans and systems and creating products that accommodate users' diverse needs, capabilities, and limitations. The designer's role is not isolated; it is interconnected with various other disciplines, particularly product management and engineering, and the organization's stakeholders, especially in delivering Software as a Service (SaaS) products.

This paper uncovers four themes to get a picture of the designer's role in creating accessible products. The first theme discusses concepts relevant to accessibility, such as universal design and inclusive design. Although these concepts have a broader scope, they share a common goal with accessibility: ensuring that a great range of people can fully experience products, services, environments, and systems. Learning about the different disabilities and their accompanying challenges gives designers insights into how they must empathize with users to create accessible and inclusive experiences. Understanding the role of organizational bodies like the World Wide Web Consortium (W3C), Web Accessibility Initiative (WAI), and standards like the Web Content Accessibility Guidelines (WCAG) to guide designers and accessibility practitioners in their process and the laws like the American Disabilities Act, Section 508, and other international regulations that ensure products are accessible is essential in gaining knowledge in the domain of accessibility. Exploring the various accessibility certifications available can help product designers further their expertise in creating inclusive digital experiences.

The second theme of this paper discusses why accessibility is a human factors concern and a core responsibility of the designer. Human-Computer Interaction (HCI), a critical discipline of Human Factors, concentrates on designing web-based systems for human use. Web accessibility is a Human Factors and HCI concern because it enables humans across all abilities to interact with the web freely and equally. Furthermore, ensuring product accessibility helps companies avoid costly lawsuits and legal penalties. Recognizing the unique challenges faced primarily by users with disabilities and organizations delivering these products and services, designers have a moral and professional obligation to create inclusive experiences that promote accessibility and equality.

The third theme of this paper uncovers accessibility gaps and challenges in product design, particularly in startups, where designers could face a lack of executive support and prioritization. Similarly, when designers juggle prioritizing work and the demands of being in a startup company, they may also lack expertise in building accessible products due to limited training and education. The lack of executive-level advocacy and awareness affects accessibility prioritization, impacting hiring, training, and investments in design, engineering, and testing tools, needing conscious efforts to ingrain accessibility practices on a team level rather than as a company-wide initiative.

The fourth theme explores further trends and possible shifts in the industry regarding accessibility, particularly with the growing prevalence of artificial intelligence. This paper examines how AI might impact different aspects of the design practice, from shifts in the design process, particularly in exploring new tools and technologies, to continuing design education on accessibility to tackle more complex user problems with the aid of AI. In closing, this article examined how companies have leverage AI to create novel methods of addressing the needs of

users with disabilities, leading to more innovative solutions and advancements in user experience and the profession altogether.

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Key Concepts

Universal and Inclusive Design

The central goal of universal design, inclusive design, and accessibility is to create products and services that cater to the needs of as many users as possible. Universal design was first coined in the article "Accessible Environments: Toward Universal Design," by architects R. L. Mace, G. J. Hardie, and J. P. Place (1991). Although the publication focused on architecture, landscape, interior, and product design for physical products, the authors describe universal design as "designing all products, buildings and exterior spaces to be usable by all people to the greatest extent possible" (Mace et al., 1991, p. 156). The authors emphasize that designers are the catalysts in enabling people with diverse abilities to participate and engage in mainstream daily activities as fully and independently as possible. Similarly, Connell et al.'s Seven Universal Design Principles aim to integrate principles such as Equitable Use, Flexibility in Use, and Simple and Intuitive Use early in the initial design process (1997). Universal and inclusive designs share some overlapping themes. Both concepts are holistic, equitable, and comprehensive, addressing barriers to all abilities in the design process and enabling user independence and equality. The concept of inclusive design emphasizes the deliberate consideration of diverse human characteristics throughout the design process. Designers must consider not just people's abilities but also diverse human characteristics, such as age, gender, ethnicity, and cultural background (Nussbaum, 2007). Accessibility tends to be more focused in

scope, concentrating on the minimum standards and accommodation for people with disabilities using products, devices, services, or environments (Imrie & Hall, 2003). These disabilities could be situational, temporary, permanent, mild, or severe. While universal design aims to deliver products, environments, and services without needing adaptation or specialized design (Mace et al., 1991), accessibility involves assistive technologies or accommodations that help adapt the experience to the user's needs. To share the clear distinction of these three concepts in a real digital product context. Here are Microsoft's definitions are mentioned in Kat Holmes, author of the book *Mismatch: How Inclusion Shapes Design* (2018).

Universal Design is defined as the design of an environment so that it might be accessed and used in the widest possible range of situations without the need for adaptation. (p. 55)

Inclusive Design: A methodology that enables and draws on the full range of human diversity. Most importantly, this means including and learning from people with a range of perspectives. (p. 54)

Learning about the breadth and overarching themes encapsulating Accessibility should be a fundamental responsibility for digital product designers. Being a responsible designer means being aware that design practice, like any other field, could be subject to bias and exclusion. Doing the job right means being aware and taking deliberate action to make inclusive and equitable digital products and services.

Understanding Disabilities

Product designers must be knowledgeable of user abilities, needs, preferences, and expectations to create solutions that allow users to accomplish their goals while using the web. In

particular, designers must be mindful of the five disabilities: auditory, cognitive, physical, speech, and visual, as noted on the Web Accessibility Initiative website. Users with an auditory disability may have hearing difficulties or are permanently deaf. Users may rely on hearing aids and sign language as their primary mode of communication, indicating a likely lack of fluency in comprehending written language. The World Health Organization suggests that an estimated 430 million people, or over five percent of the world's population, are deaf or hard of hearing. By 2050, the number could rise to at least 700 million, requiring rehabilitation. Even more alarming, the international organization predicts that nearly 2.5 billion people will experience hearing loss (Deafness and Hearing Loss, 2024).

Cognitive disabilities can be broadly categorized into clinical and functional types (Cognitive, n.d.). Clinical disabilities encompass conditions such as learning disabilities, neurodevelopmental disorders, and neurological disorders resulting from head injuries. Functional disabilities, on the other hand, refer to challenges faced in daily life due to cognitive disabilities beyond any underlying medical or behavioral causes. Memory impairments are a typical example of a functional cognitive disability. WebAIM's article on cognitive disabilities highlights that this category is the most prevalent globally (Cognitive, n.d.).

Motor, mobility, or physical disabilities may involve complete or partial paralysis, loss or damage of limb, seizures or tremors, and challenges with manual agility, coordination, and fine motor skills. These disabilities could be genetic, acquired with age, or caused by disease or an accident. Users with physical disabilities needing control devices could encounter issues controlling them when interacting with the product. As a result, users may rely on assistive technologies such as voice-activated software. Prolonged usage of these technologies may also cause fatigue.

Speech disabilities involve problems with producing speech, ranging from mild speech deficiency to permanent speaking disability. According to the learning guide for the Certified Professional in Accessibility Core Competencies test, users with this disability may experience articulation disorders such as speech sound, phonological process, or motor speech. Brain-related trauma or injuries could also result in aphasia, a language-specific impairment, or much worse, mutism, the inability to speak. (International Association of Accessibility Professionals [IAAP], 2020).

Users with visual disabilities experience permanent, temporary, or situational blindness. Blindness may involve mild to complete vision loss. Low-vision users, for example, more commonly older people, may have difficulty performing everyday activities. Eye ailments are common among older users and cannot be fully resolved by glasses or surgery. One other type of visual disability is color blindness. Users with this permanent condition have difficulty seeing the correct color or distinguishing various colors.

When considering diversity in user needs, one must be aware that disability has other layers that affect usability besides the medical conditions. These aspects could be age and health-related disabilities, multiple or changing abilities, and temporary, situational, or permanent disabilities (Diverse Abilities and Barriers, 2017).

W3C, WAI, and WCAG

The World Wide Web Consortium, or W3C, is an international organization that aims to make the Internet consistent, operable, and accessible to everyone. Tim Berners Lee, the World Wide Web and W3C founder, understood the importance of web standardization for the system's sustainability worldwide. In 1994, W3C was founded by Berners Lee at the Massachusetts Institute of Technology (MIT) with the support of colleagues from CERN, DARPA, and the

European Commission ("A Short History of the Web", n.d.). Web standards are created collaboratively by members of the World Wide Web Consortium (W3C) and guided by three core principles: universality, transparency, and consistency. ("Our Mission", n.d.). These standards evolve to reflect the changing needs of diverse users and developers.

The Web Accessibility Initiative, or WAI, is a specific program within W3C that develops guidelines and resources, such as the Web Content Accessibility Guidelines (WCAG). To help make the web accessible to people with disabilities, the WCAG follows the perceivable, operable, understandable, and robust principles (POUR). The perceivable principle ensures that users can make sense of the content, for example, by providing text alternatives for images and non-text content. The principle of operability ensures that users can interact and navigate through the site quickly, with or without using assistive technologies. One example of making the web product or site operable is to ensure that users can navigate the web entirely using the keyboard. The third principle focuses on text content and the need to make it understandable so users can avoid and overcome mistakes easily. For example, ensuring that the language used is defined in code and providing appropriate and simplified definitions to make text content easy to understand. The last principle, robust, ensures that the web product or service is compatible with an extensive list of assistive technologies, software applications, and devices used to interact with the web product or service. Providing screen readers with readable code is a primary example of this principle.

The WCAG is structured into three conformance levels: A, AA, and AAA, with each category representing an increasing level of complexity and compliance. Level A is the most basic level, while most companies and organizations with a web presence aim to achieve at least

the newest WCAG 2.3 AA conformance level. Level AAA is the most stringent and rigorous conformance level, usually applicable to industries subject to specific regulations or laws.

Product design teams aim to deliver web experiences that are functional, easy to use, and, most importantly, accessible to a vast audience. Websites and SaaS are entirely subject to government regulations and laws; being well-versed in accessibility ensures that the company is compliant, minimizing the risks of potential lawsuits. Finally, web products and services that are accessible scale better over time, as WAI ensures that the standards and guidelines adapt and evolve as regulations and user expectations.

Laws and Regulations

The W3C provides the frameworks, guidelines, and standards to make the web accessible. Still, the country's governing body upholds these standards through regulations and laws to ensure that all constituents can access the web equally. These laws remove the barriers that discriminate against people with disabilities from wholly exercising their rights in society.

In the United States, the Americans with Disabilities Act (ADA), is the primary law that covers accessibility across varying facets such as employment, telecommunications, government bodies, and public and commercial entities. This law protects web accessibility.

Section 508, an amendment to the Rehabilitation Act of 1998, focuses on upholding accessibility for federal agencies and their use of information and communication technology (ICT). Section 508 uses the WCAG as a reference standard to ensure that federal bodies comply when developing, procuring, maintaining, or using electronic and information technology (Rehabilitation Act of 1973, n.d.).

Similarly, the 21st Century Communications and Video Accessibility Act (CVAA) and the Air Carrier Access Act (ACAA) are specific laws that ensure the accessibility of

communication and video services, mainly desktop and mobile content and air travel. One crucial detail is that accessibility laws evolve through amendments and regulations.

Outside of the United States, the United Nations Convention on the Rights of Persons with Disabilities (CRPD) is a treaty that defines the rights of people with disabilities, and the European Accessibility Act is a law that captures similar requirements as the ADA for products and services, including digital applications.

Accessibility Certifications

Product designers can become accessibility experts through multiple paths. One path is through experience and learning from accessibility mentors within the company. Another path is to participate in online or in-person seminars and classes. Certification could be a legitimate option to become an accessibility expert.

The International Association of Accessibility Professionals (IAAP) offers five types of certifications (Certification, n.d.). The Certified Professional in Accessibility Core Competencies (CPACC) is an introductory examination that provides a foundational yet comprehensive accessibility test across different domains and devices. Another certification for the same organization is the Web Accessibility Specialist (WAS). This technical proficiency certification certifies the candidate as having hands-on experience in testing, evaluating, and implementing web products. A level higher than CPACC and WAS is the Certified Professional in Web Accessibility (CPWA). Both the CPACC and WAS are prerequisites for this exam.

The Certified Professional in Accessible Built Environments (CPABE) is higher than CPACC and allows candidates to deepen their accessibility knowledge. This certification offers multiple levels, such as associate, advanced, or expert-level accessibility professionals. Lastly, the IAAP Accessible Document Specialist (ADS) is a technical certification for candidates

interested in creating and evaluating accessible electronic documents such as word processors, spreadsheets, and presentations. Certified professionals must maintain these certification requirements by gaining continuing credit that can be completed every three years.

Why is Web Accessibility a Human Factors Concern, Therefore, a Designer's Concern?

Human Factors is about understanding the relationship between humans and the systems surrounding them. In the seminal book "Engineering Psychology and Human Performance," Wickens, Hollands, Banbury, and Parasuraman (2015) emphasize the importance of knowledge of human behavior and its direct influence on the design of products, systems, and environments. Human-Computer Interaction (HCI) is one of the critical disciplines under the Human Factors umbrella that concentrates on designing, evaluating, and implementing web-based systems for human use (Hewett et al., 1992). Web accessibility is an HCI concern because it primarily enables humans across all abilities to interact with the web freely and equally. To genuinely empathize with users and understand how they interact with technology, product designers must be able to gain insights and provide innovative solutions by examining users' needs, challenges, goals, and behaviors.

These aspects of the user vary dramatically based on their abilities to interact with the web. By recognizing the unique challenges and constraints faced by users with disabilities, designers have a moral and professional obligation to create meaningful and inclusive experiences that promote accessibility and equality. According to the World Health Organization (WHO), roughly 16% of the global population or 1.3 billion people, experience significant disability. People with disability face various forms of inequities directly from the systems and environments that are supposed to provide them equal opportunities and rights. Unfortunately, people with disability are more likely not to use the Internet or even own a device. The survey

results done by the PEW Research Center in 2021 suggest that only 62% of U.S. adults with a disability own a computer device, compared with 81% of participants without disability (Pew Research Center, 2021). Disability could be compounded forms of marginalization. This compounding effect could be similar to psychologist Kimberlé Crenshaw's concept of intersectionality, where individuals can experience multiple, overlapping forms of discrimination because of their identity and socio-economic background (Crenshaw, 1989).

By ensuring the product is accessible, designers make the product legally compliant, mitigating the chances of costly lawsuits and legal penalties. In an article posted by Kristina Launey and Minh Vu on the Seyfarth Shaw LLP website, over 2,500 ADA website accessibility lawsuits were filed in federal courts in 2020, an abrupt 12% increase from 2019 (2021). One high-profile case was the lawsuit by the U.S. Department of Justice against edX Inc., a well-known online learning platform co-founded by Harvard University and the Massachusetts Institute of Technology (MIT). A four-year agreement was settled, requiring the online platform edX to make significant accessibility improvements to its digital and mobile experiences. This settlement was in conjunction with a similar lawsuit by the National Association of the Deaf against both universities, claiming that they have discriminated against deaf and hard-of-hearing individuals by failing to provide captions for their online courses (2015).

When accessibility is considered and evaluated throughout the design and development process, the company is legally compliant, and the web product or service could have the potential for a truly differentiated experience.

Accessibility Gaps and Challenges in Product Design

According to the 2023 Year End Report entitled ADA Digital Accessibility Lawsuit Trends Involving: Websites, Mobile, or Video by UsableNet, more than 4,600 cases were filed in

federal and state courts last year. The e-commerce, food service, and education industries receive the most lawsuits. The data gathered by UsableNet suggests that law firms typically target businesses with less than 25 million annual revenue and pursue these cases en masse. What's even more staggering is that 25% of this total number were lawsuits against businesses that were previous offenders, targeting primarily desktop websites followed by all the other application interfaces. This overwhelming statistic suggests much must be done to ensure the web is accessible. Achieving digital accessibility requires a coordinated effort across multiple teams and roles. The design team, product, and engineering play a pivotal role in embedding accessibility into the product development cycle and becoming advocates to help institutionalize it as a critical differentiator.

Lack of Executive Support and Prioritization

Lack of prioritization is a critique mainly of businesses and organizations in which the design teams are embedded, particularly the lack of initiative-driven top-down from the executive level to make Accessibility a key product pillar. According to the 2016 Forrester Consulting Impact Study sponsored by Microsoft, to understand the value of investing in accessible technologies, the key barrier is the perceived belief that Accessibility is costly. 31% of those 319 business and technology leaders from Europe, the reason for the high cost of establishing and maintaining accessibility programs and the lack of funding, and some thought the return on investment was lacking. Only 17% of those leaders thought their accessibility initiative was highly successful. In the article "The Accessibility Business Case for IT" by Info-Tech Research Group, an information technology research and advisory company, Executive buy-in is challenging, especially if the organization lacks an understanding of how Accessibility fits into the overall business strategy (2023). Lack of executive support means accessibility

advocates such as product designers have put in extra effort to communicate the value and benefits of Accessibility, particularly in creating an accessibility program and incorporating it into the product strategy (*How to Hire for Digital Accessibility Roles — Smashing Magazine*, 2022). Most often, Accessibility becomes an afterthought.

Furthermore, inaction to fix and improve the product through the lens of Accessibility could exponentially increase unplanned costs. Inevitably, companies must invest in tools, technologies, and training to support accessibility advocates to avoid lawsuits and be competitive. According to the WebAim article “The Business Case for Digital Accessibility,” accessible products can extend market reach (2018), with over 3.3 billion potential consumers, including family and friends of people with disability (Return On Disability Annual Report, 2020). Competitors that prioritize Accessibility deliver better user experiences and may even uncover untapped differentiating opportunities. The benefits outweigh the perceived cost because accessible products not only aid people with disabilities but also account for diverse needs and situations faced by users without disability. If designing accessible products is an organization-wide mission, every project gets the concerted effort, visibility, and prioritization it deserves, empowering designers, including the overall product and engineering teams, to put the users first.

Prioritizing accessibility benefits customers, the company, and employees. Accessibility goes beyond ensuring the web product is inclusive. Organizations must think of the overall environment, including the employee experience. As mentioned in the 2016 Forrester Consulting Impact Study, organizations must consider what truly makes employees engaged and productive, especially for employees with disabilities. The benefits of creating an inclusive organization include:

- Increased productivity.
- High employee retention.
- Elimination of potential legal fees.
- Improved morale.
- More comprehensive access to the talent pool.

Prioritizing Accessibility could raise customer brand loyalty and employee trust.

Limited Resources and Training

The accessibility gap of limited resources and expertise could affect the design team for several reasons. For example, early-stage startups earning yearly revenue of \$25 million (UsableNet, 2023) have fewer employees than growth-stage companies, with a more substantial likelihood of lacking dedicated accessibility roles. Furthermore, companies with minimal accessibility expertise often struggle to source talent for accessibility-specific functions (*How to Hire for Digital Accessibility Roles — Smashing Magazine*, 2022). Also, with limited resources and funding constraints, accessibility can take a backseat or become an “add-on” compared to other priorities, especially without solid accessibility champions in leadership. The survey done by Skynova, an invoicing software company, entitled “Why startups failed in 2022”, found that lack of financing and investors and running out of funding are the top two obstacles to startup success. To rapidly grow and scale operations, startups must be able to pivot or quickly change their business strategy by either starting over, changing the core features, functionality, or even underlying technology. Another critical characteristic of pivoting is the ability to change course before running out of resources (Founder Institute, 2023). When struggling with limited funds and contested feature roadmaps, prioritizing accessibility can seem daunting. Startups often focus on core product development, such as the “minimum viable product” (MVP) and achieving

initial traction to drive growth. With top-level directives to focus on features, designers lacking knowledge and training on accessibility could, unfortunately, establish a culture that bypasses accessibility altogether.

Some startups specifically cater to the needs and challenges of people with disabilities and have received funding from sought-after investors (Magistretti, 2023) and grants from tech giants like Microsoft (Coldewey, 2019). A couple of essential factors are apparent across these companies. One is that executives are accessibility champions, with some born with disabilities like Founder John Robinson of Our Ability Inc. (*Team – Our Ability, Inc.*, n.d.) or have relatives with disabilities like Pison Founder Dexter Ang (Pison Technology, 2024). The second is that the target users are people with disabilities. The most recent analysis by Mordor Intelligence suggests that the digital accessibility software market size is estimated to reach \$958.7 Million by 2029, with North America taking the largest share in the market for digital accessibility software in 2024 (*Digital Accessibility Software Market Size & Share Analysis- Growth Trends & Forecasts (2024-2029)*, n.d.).

These accessibility challenges explicitly faced by product design teams are all interconnected. The lack of advocates and awareness on an executive level affects accessibility prioritization, which then permeates across multiple facets in how hiring, training, and investments around design and engineering practices and testing tools—requiring conscious efforts to ingrain accessibility practices on a team-based scale instead of as a concerted company initiative. In closing, further research studies on the correlation between startup success and investments in accessibility are needed.

Possible Future Trends

The Design Practice

External and internal demands for companies to prioritize accessibility will significantly enhance the design practice. Leading the way in creating more inclusive systems and processes, big tech companies like IBM, Microsoft, Apple, and Google have developed tools, toolkits, guidelines, and research initiatives focused on accessibility. With tested credibility and brand image, these companies often lead the way in adopting new technologies or practices—focus on accessibility is one of them.

For example, Microsoft, along with its robust online accessibility platform, hosted its second annual Ability Summit, a free, one-day digital event held on March 7, 2024 (Ability Summit 2024, 2024). The summit featured individuals with disabilities, allies, and accessibility leaders who shared valuable insights. Discussions explored the potential of AI in driving accessibility innovation and bridging the accessibility gap. Similarly, axe-con 2024, the most prominent digital accessibility conference, occurred earlier this year (axe-con 2024, n.d.). This event welcomed a global audience of developers, designers, business leaders, and accessibility professionals of all experience levels, focusing on building, testing, and maintaining accessible digital experiences. Both events offer free registration and can be accessed online.

As continued visibility on accessibility grows, pioneered by widely used products and services, access to training and education becomes increasingly inexpensive and readily available. As market demands for roles with strong accessibility backgrounds grow, designing products specifically for people with disabilities or mass access, designers must prioritize gaining knowledge in creating and communicating inclusive designs. For example, on the traditional design education front, universities and colleges across the globe would increasingly offer Accessibility courses as part of the design and engineering curriculum. The University of Washington (*Accessibility-Related UW Courses*, n.d.), Carnegie Mellon University (*Special*

Topics: Accessibility: A Guide to Building Future User Interfaces | Human-Computer

Interaction Institute, n.d.), and Stanford (*Accessibility Training Modules*, 2024), to name a few, offer accessibility-specific courses. The Internet also offers a plethora of reliable paid and free courses online. Increasing focus on accessibility education and training, such as mentorship from an accessibility expert or even certifications to further professional development, will be a trend designers will continue to pursue.

The increasing market demand for accessible digital products and services, coupled with a growing focus on accessibility, will likely encourage more designers with disabilities to apply for design jobs. As companies prioritize accessibility, they are more likely to adopt inclusive hiring practices that actively seek out and recruit designers with disabilities. Designers with disabilities bring unique perspectives and provide valuable insights into the challenges and needs of users with disabilities, leading to more empathetic and inclusive design solutions and processes. Most importantly, seeing the trend of increased representation of designers with disability can help break down stereotypes and create a more diverse and inclusive community.

Design Tools

Continued focus on efficiency and scale to better incorporate accessibility into the design process would be an essential trend for businesses in a growing competitive market of an estimated 1.3 billion people (World Health Organization: WHO, 2023). Figma, the company providing a well-known cloud-based design prototype tool for designers, emphasizes creating more inclusive systems (*The Future of Design Systems Is Accessible | Figma Blog*, 2023).

Continued investments in design systems, a library of guidelines, principles, UI components, and best practices that help teams design and develop digital products consistently and efficiently can catalyze accessibility, enabling organizations to bake in inclusive design principles into their

product design and development process at an unprecedented scale. This trend would intrinsically require designers to be knowledgeable in designing and testing for accessibility, and as a result, could exponentially drive efficiency and thorough compliance as design systems streamline the process of implementing widespread change through components, styles, and variants that can effortlessly change all linked instances.

Being able to validate and test designs for accessibility with the aid of responsible accessibility tools, particularly the ones driven by AI, could help automate specific steps in the design process, from checking color, text readability and legibility, focus orders, and many more (Vjalihs, 2022). Design knowledge of industry-wide accessibility guidelines and best practices is critical, especially when troubleshooting issues requiring human intervention. Like training and education, tools and technologies used by designers will become increasingly open-source and accessible (Brose, 2022). Open-source tools pave the way for helping designers and engineers improve their products and staying true to the goal of accessibility — ensuring a future where everyone can contribute to the technology usable by as many people as possible.

The Artificial Intelligence Wave in Accessibility

Artificial intelligence (AI) is poised to redefine the boundaries for web products and services in science, business, healthcare, education, e-commerce, and many more, opening up more opportunities for people with disability to interact with the web. Some believe AI could be the great equalizer, democratizing access to the web (van Loon, 2023). With diverse and innovative solutions to bridge the gap, from predictive AI and brain-computer interfaces to further enhancing widely accepted assistive technologies, design is central to creating human-centered and inclusive web products and services that could expand globally, reaching vulnerable and underprivileged users with disability. Furthermore, with the growing demands

and competitive market, AI-powered accessibility tools must focus on multi-modality, function, personalization, and delight. Products that offer alternative options to access information, personalized options based on their specific needs, preferences, and goals, and promote independence will have a higher chance of succeeding in this era of AI.

Visual Disabilities

With AI, companies can explore innovative ways to offer more personalized and tailored experiences for the unique challenges of people with visual disability. For example, startups like Typehead AI leverage natural language commands instead of relying solely on a keyboard or alternative keyboard devices, further expanding on the current limitations of screen readers. When done right, context and intent recognition through natural language commands could lead to delightful and authentic user experiences. AI could enhance the experience by recognizing the person's preferences, such as reading speed, voice style, and content interests.

Furthermore, the growing success of generative AI could be leveraged in content summarization, making it easier for screen readers to summarize and absorb long-form content. Computer vision models could also deliver more accurate and detailed descriptions of images and videos. For example, the Danish mobile app Be My Eyes offers real-time visual assistance to people with visual disability through an AI that provides detailed and context-specific descriptions of scenes and images (Costabel, 2023). Even ChatGPT allows blind and low-vision users to hear image descriptions (Brownell, 2024).

Auditory Disabilities

Video and audio content can be further accessible through AI. Real-time, accurate, and contextually relevant processing of captioning and transcription through Natural Language Processing (NLP) could significantly aid users with hearing disabilities or users who are not

proficient in the subject language. With machine learning algorithms' ability to ingest and learn extensive data at an unimaginable speed, recognizing speech patterns, accents, and intonation, AI can anticipate intent, making reading captions live, allowing users with hearing disabilities to comprehend the information simultaneously with those without disability. Companies like Assembly AI leverage Automatic Speech Recognition (ASR) and can effectively transcribe speech-to-text content for various use cases such as content management, video hosting, and live streaming, which could greatly help customers requiring real-time transcriptions for hearing-impaired audiences. Sound recognition AI could detect and classify noise from important sounds to provide visual or haptic alerts to impaired hearing users in crucial situations. AI-powered translation tools like Meta's open-source AI model, SeamlessM4T (Meta, 2023), can translate and transcribe into over 100 languages. Similarly, Google Cloud's Speech API (Barnes, 2023) supports over 125 languages and variants, making web content more accessible to users in various sign or written languages.

Motor, Speech, and Cognitive Disabilities

AI has the potential to revolutionize communication for individuals with motor, cognitive, and speech disabilities by leveraging the power of predictive text and voice recognition. These technologies can analyze individual typing and speech patterns, allowing the AI to predict and recommend relevant words, expressions, and even complete sentences; thereby, typing messages, composing emails, or engaging with the web could be way different from now. For example, Voiceitt leverages AI to break down communication barriers by recognizing and translating speech from individuals whose verbal expressions are complex to comprehend due to medical conditions, comorbidities, or age-related circumstances.

With Amazon's Alexa, Apple's Siri, Google's Assistant, and recently the Meta AI Assistant paving the way in the mass market, voice technologies powered by AI will become even more sophisticated and ubiquitous in our day-to-day lives, opening more innovative opportunities to uncover solutions to daily problems that have not been solved yet. The multi-modal nature of these assistants enables users with different abilities, particularly motor and cognitive, to choose their preferred mode of communication—meeting where they are. Multi-modal applications can also aid users in a human-like interaction using a combination of voice, text, and even visual elements. This innovation can be particularly beneficial for users with cognitive or social disabilities. For example, Google recently launched Gemini, its new AI-powered assistant that can be accessed over chat, voice, and haptic feedback if you access Google's Pixel Watch. This chatbot is built-in with almost all Google devices, like having your personal assistant (*Google Assistant, Your Own Personal Google*, n.d.).

Conclusion

To effectively address the product designer's role in making products accessible, a systematic approach must be taken to consider the processes, individuals, training, tools, and technologies involved. An accessibility-centered design process ensures that the needs and challenges of user disabilities are established as fundamental aspects of the product's design. Working with fellow designers, engineers, and members of the broader company should reinforce a shared understanding of what accessibility means and its importance, and most importantly, everyone's role in ensuring the product's success. Every employee, particularly designers, should be encouraged to adopt inclusive design practices throughout the company. Design practices should manifest in communicating accessibility goals and decisions, facilitating research, gathering feedback from users with disabilities, and documenting requirements. Hence,

the design is clearly and sufficiently translated for engineers to turn it into compliant code. The increasing demand for fully accessibility web products and services results in increasingly sought-after accessibility-specific design expertise in the job market. Establishing and standardizing education and training on accessibility for design practitioners could radically prepare and equip beginner and tenured designers. The role definition between designer and accessibility expert becomes integrated, and previously mentioned gaps in early-stage startups become less apparent.

Lastly, with the prevalence of AI-powered tools and technologies, the gaps and challenges in building accessible experiences become narrower. With the aid of AI, product designers could potentially see new ways of solving customer problems—innovating and furthering the profession.

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